Submission guidelines

1. Project Report:
   Submit a well-structured report that includes a clear objective, problem statement, and a thorough explanation of the approach taken to solve the problem.

2. Code and Explanation:
   Include the complete source code along with detailed explanations of each major section or function, describing what each part of the code is responsible for.

3. Results and Discussion:
   Present and discuss key results, highlighting significant observations, findings, and any challenges faced during development.

4. Presentation Slides:
   Prepare a concise and informative presentation that summarizes the project, methodology, key outcomes, and conclusions.

Automated Class Routine Generation System Using C Programming

Design and implement an automated class routine (timetable) generation system using programming language. The objective of this project is to simulate a real-world academic scheduling problem and develop a robust, and flexible solution using structured programming concepts. The program should allow a user (acting as an academic coordinator) to input and manage institutional resources such as classrooms, teachers, courses, and student sections, and then automatically generate feasible class routines based on given constraints and priorities.

Functional Requirements:

1. The system must support adding any number of classrooms, each having a full name, short name/code, capacity, and type (e.g., theory lab, computer lab, general classroom).
2. The system must allow adding any number of teachers, where each teacher has:

    a) Full name and short name
    b) Seniority level (e.g., numeric rank)
    c) Maximum teaching load per day/week
    d) Preferred time slots.

3. The system must allow inclusion of any number of courses, each having:

    a) Course title and short code
    b) Credit hours
    c) Required class duration (1-hour, 2-hour, or 3-hour slots)
    d) Required classroom type (if applicable)

4. The daily academic time window is 8:00 AM to 5:00 PM, which must be divisible into 1-hour, 2-hour, or 3-hour slots, depending on course requirements.
5. The system must support adding multiple sections (classes) of students, where each section has:

    a) Section name/code

b) Number of students
c) Assigned home classroom

6. Each section must have a home classroom, but the system must allow assigning a different classroom for specific courses if required (e.g., labs or shared facilities).
7. If needed, a section must be dividable into multiple subsections, and the routine should handle them independently while still avoiding conflicts.
8. The program must automatically detect and report conflicts:

    a) Two sections assigned to the same classroom at the same time
    b) A teacher assigned to multiple classes at the same time
    c) A section scheduled for overlapping classes
    d) Classroom capacity being insufficient for assigned sections

9. Teacher seniority must be prioritized during routine generation; such that senior teachers are allocated their preferred or optimal time slots before junior teachers.
10. Each teacher rank (Professor, Associate Prof, Assistant Prof, Lecturer) has a predefined maximum teaching load per week, which must be enforced during routine generation.
11. The system must generate separate timetables preview for:

    a) Each section
    b) Each teacher
    c) Each classroom
    d) Each course

12. The system must support both full names and short names (codes) for teachers, courses, classrooms, and sections, and use them appropriately in outputs.
13. All data (teachers, courses, classrooms, sections, routines) must be stored and retrieved using file handling so that the program supports multiple executions without data loss.
14. The final routine output must be exportable in a printable format (e.g., pdf).

Sample output:

Final Routine (Term: July 2025)

Muhammad Abdullah

	1st 8:00 - 8:50	2nd 9:00 - 9:50	3rd 10:00 - 10:50	4th 11:00 - 11:50	5th 12:00 - 12:50	6th 13:00 - 13:30	Break 1:30 - 2:00	7th 2:00 - 2:50	8th 3:00 - 3:50	9th 4:00 - 4:50
Saturday				IPE 2/1 B ME 245 IPE 402	IPE 2/1 A ME 245 IPE 401			ME 3/2 A FM Lab ME 326 TH / SA / MIK / MAb / TA-2		
Sunday				ME 1/2 C UCL-1 ME 172 Group 1 MAb				ME 3/2 C R#ME 405 ME 352 Group 2 ARMA / MAb		
Monday										
Tuesday		IPE 2/1 B ME 245 IPE 402	IPE 2/1 A ME 245 IPE 401	CSE 1/2 B Drg Lab (R# 416),UCL-1,UCL-2 ME 174 (CSE) MAb / DI / TA-3				IPE 2/2 B AM Lab,SM Lab ME 244 (IPE) JAK / AS / SA / MAb / TA-3		
Wednesday				ME 2/2 B AM Lab,SM Lab ME 244 RHR / MAb / MHR / MSA / TA-4						

Timetable generated:12/10/2025 aSc Timetables

Final Routine (Term: July 2025)

ME 172

	1st 8:00 - 8:50	2nd 9:00 - 9:50	3rd 10:00 - 10:50	4th 11:00 - 11:50	5th 12:00 - 12:50	6th 13:00 - 13:30	Break 1:30 - 2:00	7th 2:00 - 2:50	8th 3:00 - 3:50	9th 4:00 - 4:50
Sa										
Su				6 MAb, SAf, TA-7 ME 1/2 C	6 MAb, SAf, TA-7 ME 1/2 C	6 MAb, SAf, TA-7 ME 1/2 C		6 MIK, TA-5, SAf, TA-7 ME 1/2 B	6 MIK, TA-5, SAf, TA-7 ME 1/2 B	6 MIK, TA-5, SAf, TA-7 ME 1/2 B
Mo										
Tu								6 SaS, TA-6, SAf, TA-7 ME 1/2 A	6 SaS, TA-6, SAf, TA-7 ME 1/2 A	6 SaS, TA-6, SAf, TA-7 ME 1/2 A
We										

Timetable generated:12/10/2025 aSc Timetables

Final Routine (Term: July 2025)

ME L-1/T-2 (Sec C)

R#ME 512

	1st 8:00 - 8:50	2nd 9:00 - 9:50	3rd 10:00 - 10:50	4th 11:00 - 11:50	5th 12:00 - 12:50	6th 13:00 - 13:30	Break 1:30 - 2:00	7th 2:00 - 2:50	8th 3:00 - 3:50	9th 4:00 - 4:50
Sa	Hum 103 R#ME 509,R#ME 510	Math 163	ME 171 MM	Hum 101 R#ME 509,R#ME 510	Phy 159					
Su	Math 163	ME 171 SMS		ME 172 UCL-1 Group 1 MAb ME 172 UCL-2 Group 2 SAf / TA-7						
Mo	Math 163		Hum 103 R#ME 509,R#ME 510	Chem 141				Shop 170 FW Shop		
Tu	Hum 101 R#ME 509,R#ME 510			Phy 159	Chem 141					
We	Hum 101 R#ME 512,R#ME 503 Hum 103 R#ME 509,R#ME 510	Phy 159	Chem 141	ME 171 MM	Math 163			PHY 102 Physics Lab		

Timetable generated:12/10/2025

aSc Timetables

Final Routine (Term: July 2025)

R#ME 405

	1st 8:00 - 8:50	2nd 9:00 - 9:50	3rd 10:00 - 10:50	4th 11:00 - 11:50	5th 12:00 - 12:50	6th 13:00 - 13:30	Break 1:30 - 2:00	7th 2:00 - 2:50	8th 3:00 - 3:50	9th 4:00 - 4:50
Sa		IPE 381 ME 3/2 B	ME 307 ME 3/2 B AKMM	ME 351 ME 3/2 B MAR	ME 323 ME 3/2 B MA					
Su			ME 351 ME 3/2 B MAM	ME 307 ME 3/2 B MAHM	IPE 381 ME 3/2 B			ME 352 ME 3/2 C Group 2 ARMA / MAb		
Mo	HUM 201 ME 3/2 A/ ME 3/2 B/ ME 3/2 C		IPE 381 ME 3/2 B	ME 307 ME 3/2 B AKMM	ME 323 ME 3/2 B TH			ME 352 ME 3/2 A Group 2 MAM / THS		
Tu	HUM 201 ME 3/2 A/ ME 3/2 B/ ME 3/2 C	ME 351 ME 3/2 B MAM	ME 323 ME 3/2 B MMR	ME 352 ME 3/2 B Group 1 THS				ME 352 ME 3/2 C Group 1 ARMA / MSS		
We		ME 351 ME 3/2 B MAR	HUM 201 ME 3/2 A/ ME 3/2 B/ ME 3/2 C	ME 352 ME 3/2 B Group 2 SaS				ME 352 ME 3/2 A Group 1 ARMA / MSS		

Timetable generated:11/25/2025

aSc Timetables